What a PhD student does in their free time: Evaluating Google AI Overview's recommendation for substituting a Japanese milk bread recipe with whole-wheat spelt flour and sourdough starter discard

Jan C. L. Lau

November 1, 2026

Disclaimer: This paper is not peer-reviewed nor submitted to any journals/conferences. This paper was written for entertainment purposes only and was not done to scientific rigor. Also, my intention of using AI in this context is NOT to replace bakers, but rather to minimize the chances of a failed baking session that would lead to food waste. For actual nutrition advice, please consult a medical professional.

Abstract

Baking has been a necessary skill for survival, though it has gradually become a popular hobby amongst PhD students worldwide. The Japanese milk bread is a soft and fluffy bread that is adored by many, though the original recipe only uses all-purpose flour. Studies have shown that the rise in blood glucose levels caused by refined flour could cause diabetic complications, hence there is a need for substitution using whole-wheat flour. However, it becomes a challenge, since taste and texture cannot be compromised. In this paper, Google AI Overview is consulted for recipe modifications, since the author (who is not a professional baker) wants to minimize food waste from baking failure. Results show that it is possible to use a 50/50 mix of whole-wheat spelt and type 550 wheat flour, as well as a 7-day-old sourdough starter discard, to bake very soft and fluffy Japanese milk bread.

Introduction

With the earliest evidence of baking dating back to more than 14000 years [1], it is evident that humans and baked goods have been inseparable. Baking methods have progressed throughout the years: the ancient Egyptians discovered and implemented natural fermentation in their baking, whereas the Greeks and Romans introduced communal ovens as an improvement on baking techniques [1]. In the "old days", baking is considered a survival skill, since it is crucial for creating food to keep humans well-fed and alive. However, as the world progresses through the centuries, industrial-grade machines and ovens have been developed to accommodate mass production of baked goods. Nowadays, chain-store bakeries sell bread at a relatively affordable price compared to other essential foods. For individuals that prefer authenticity over mass production, one can also purchase artisanal bread at artisanal bakeries at a higher price [2].

Given the abundance of readily-available bread and coupled with busy lifestyles of individuals worldwide, the need to acquire baking skills for survival reasons gradually fades. However, baking can surprisingly be a popular hobby among PhD students for different reasons, ranging from wanting to escape burnout and endless academic stress [3, 4] to enjoying tangible results through control and predictability [5].

The author of this paper has been living in Karlsruhe, Germany for 3+ years. While Germany is famous for their amazing bread, of which the author enjoys, there are no bakeries in the city to the author's knowledge that bakes Asian-style soft and fluffy milk bread to satisfy the occasional cravings. Furthermore, the author is also a PhD student who is actively looking for different injury-free methods to relieve stress. As such, a google search was conducted on soft fluffy milk bread recipes. The "Japanese Milk Bread (Hokkaido)" recipe by Jacky Kwok was found and used [6]. Initial executions of his recipe showed great success across taste, smell, and texture.

Blood glucose refers to the amount of sugar present in a person's blood at any time [7], with postprandial glucose (PPG) defined as the concentration of plasma glucose in the body after eating [8]. The glycemic index (GI) is a value that classifies carbohydrates based on the increase in blood glucose levels after consumption with respect to a reference food (which can be glucose or white bread) [9], and it is said to be one of the factors that can influence PPG [10]. As summarized in [10], the American Diabetes Association declared that PPG contributes to diabetes complications [8] and another source stated that the increase in PPG can cause increase in insulin after meals [11]. In rice, it was found that brown rice produces lower PPG and postprandial insulin responses than white rice [12]. In bread, it is shown that bread containing whole-meal spelt flour has lower GI values than white spelt bread and white-wheat-flour bread [13]. Another study also showed that bread with sourdough starter and yeast yields slightly-lower blood glucose levels compared to bread with only yeast [14].

With the original recipe based only on all-purpose (AP) flour, a need exists on modifying Kwok's recipe to lower glucose index without compromising taste and texture. Therefore, the purpose of this paper is to 1) investigate options of lowering GI index of an original milk bread recipe by substituting whole-wheat spelt flour, 2) investigate the possibility of incorporating sourdough starter discard to minimize waste, 3) evaluate Google AI Overview's recommendations for incorporating whole-wheat spelt flour and sourdough starter discard.

Methods

Recipe Modification

To maximize success and minimize waste that could yield from failed baked goods, the author, who is not a professional baker, asked Google AI Overview for substitution recommendations using the following prompt: "*The original recipe for Japanese milk bread requires Tangzhong (2 tbsp water, 2 tbsp AP flour, and 4.5 tbsp milk), 380 g AP flour, 130 g milk, a pinch of salt, 60g sugar, 1 large egg, 10g yeast, and 43g unsalted butter, how should the modified recipe look like if I want to incorporate some 7-day old sourdough starter discard and whole-wheat spelt flour? Please also provide the flour names in German.*"

The Google AI Overview responded with the following recipe adjustments and methodological recommendations:

- *Tangzhong* (2 tbsp water, 2 tbsp AP flour, 4.5 tbsp milk)
- 100 g sourdough discard (straight from the jar, unfed)
- 165g whole wheat spelt flour
- 165g Weizenmehl Type 550 flour
- 105g milk
- 14-15g instant yeast
- 60g sugar
- 43g butter
- 1 large egg
- A pinch of salt

Experimental Procedure

The *Tangzhong* was first cooked in a 2-L stainless steel pot (IKEA 365+, IKEA), which was constantly mixed under low heat using a silicone spatula. While the *Tangzhong* was cooking, the ingredients were weighed using a standard kitchen scale. The *Tangzhong* is considered cooked when the mixture can be scooped in one go without dripping off the spatula. All ingredients except the butter were first added in a 26cm-diameter plastic bowl with cover (GARNITYREN, IKEA). Once the ingredients were somewhat thoroughly mixed, the butter (softened with a microwave) was added into the dough.

The dough was then proofed in the same plastic bowl with a cover for one hour under a 22.8˚C environment. After one hour of proofing, the dough was divided into roughly 12 equal pieces (determined per visual and tactile inspections) and placed in an oiled 3-L glass baking tray (IKEA 365+,

IKEA) in a 3x4 configuration. The divided dough underwent a second round of proofing for 20 minutes in the baking tray, covered. During the second round of proofing, the oven was preheated to 175˚C.

After the second round of proofing, egg wash was applied to the top of each piece, followed by sprinkles of black sesame to the author's desire. Once the oven was preheated to 175˚C, the glass baking tray with the raw bread was placed in the oven to bake for 20 minutes.

Variables Analyzed

Size, taste, texture, deflation, and smell were analyzed. The size of the bread was measured using a banana and measuring tape. The "banana for scale" was implemented since it is a popular unit of measure on the Internet [15], whereas the measuring tape was used for exact measurements. While the smell test was conducted immediately once the baking was complete, the size, taste, texture, and deflation tests were conducted after 45 minutes of cooling. This is to avoid skin and gastrointestinal lesions resulting from thermal exposure.



Figure 1. Top and side views of modified milk bread, with a banana for scale.



Figure 2. A deflation test is conducted to evaluate the softness of the bread.

Results and Discussion

Upon 20 minutes of baking at 175˚C, the entire apartment smelled like freshly-baked milk bread. As shown in Figure 1, the bread looks beautiful with a crispy top.

Each bread measured to be an average of 6cm (width) x 6cm (depth) x 7cm (height). The texture was softer than the original recipe, with the deflation test showing a compression from 7cm to 2.5cm (see Figure 2), which maps to a 64.3% decrease in height. As for the taste, a pleasant fermentation note was present, the black sesame addition brought a secondary flavour to the milk bread, and it tasted sweeter than the original recipe despite keeping the same amount of sugar (60g). These phenomena could be caused by the inclusion of the sourdough starter discard.

Despite the many successes of the modified recipe, this study still presents limitations. The evaluation described above was only conducted on one out of the 12 pieces due to spatial limitation of the stomach. One recommendation is to recruit additional participants and using a Likert scale-based survey to record feedback. Regarding the deflation test, the results can only be considered in a qualitative manner, since the axial force applied should also be measured. Therefore, it is recommended to conduct the deflation test on a kitchen scale. To enable comparable data with the original recipe, it is recommended to also conduct another experiment with the original recipe to obtain the same measurements described. Since the author is not a nutritionist, it is unknown how much of the GI is reduced with the 50/50 split of whole-wheat spelt and

type 550 flours. Nevertheless, this experiment opens new opportunities, including and not limited to investigating optimal spread and beverage pairings for breakfast, lunch, snack, and dinner.

Conclusion

The results presented show that the substitutions recommended by Google AI Overview did not compromise taste and texture, and in fact improved the original recipe. Therefore, it is possible to bake soft and fluffy milk bread using whole-wheat spelt flour and sourdough starter discard.

Conflict of Interest

The author declares no conflict of interest.

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